

A Charge-Pump PLL on Tiny Tapeout GF180

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Abstract

We propose a classic type-II charge-pump phase-locked loop (PLL) for the Tiny Tapeout GF180 shuttle. A phase-frequency detector (PFD), a $\div 8$ toggle-flop prescaler, a programmable feedback divider and a lock detector are built from standard cells; a charge pump, a passive loop filter and a current-starved ring VCO are analog. The loop multiplies a 5 to 12.5 MHz reference by $PN = 8$ to 40 ($P = 8$, $N = 1$ to 5) onto a VCO range of about 50 to 200 MHz (*est.*), with a nominal point of $6.25 \text{ MHz} \times 16 = 100 \text{ MHz}$ and a lock time under $50 \mu\text{s}$ (*est.*). Only the prescaler runs above 50 MHz. Digital blocks and the full loop are verified in CocoTB against a real-number model; the analog blocks are characterised in ngspice across corners. An alternative implementation moves the pump output to a tri-state pad and the filter and VCO off chip.

1 Introduction and motivation

The charge-pump PLL is the standard clock multiplier in integrated circuits, and it is the one circuit in which a digital phase detector, an analog integrator and an oscillator form a single feedback loop. Building one exercises the whole mixed-signal flow: RTL and CocoTB for the digital half, transistor-level design and ngspice for the analog half, and layout checked with DRC and LVS. The loop multiplies the tile reference by a programmable ratio, so the result is directly observable on the tile's digital outputs.

The design targets a two-tile-high analog tile built on `ttgf-analog-template` [5], with two analog pins (`ua[0]`, `ua[1]`). Section 3 describes the design in its analog-tile form, and Section 3.2 gives an alternative implementation on the standard digital tile.

2 Target specifications

These are hand estimates; replacing them with ngspice results is the Nov 5 deliverable. Only the prescaler runs above 50 MHz (Section 2.1).

TABLE 1 *the targets and where each one comes from*

PARAMETER	TARGET	BASIS
Reference	clk, 5 to 12.5 MHz; nominal $6.25 \text{ MHz} \times 16 = 100 \text{ MHz}$	well inside the 50 MHz tile clock limit
Prescaler	fixed $\div 8$ ($P = 8$), three toggle flops	smallest power of two taking the fast-corner 333 MHz to 50 MHz or below
N	1 to 5 (3 bits); $PN = 8$ to 40	lock needs $50 \leq 8N f_{\text{ref}} \leq 200 \text{ MHz}$: $N = 5$ at 5 MHz, $N = 1$ from 6.25 MHz; output steps of $8f_{\text{ref}}$
VCO range	50 to 200 MHz; 333 MHz fast corner (<i>est.</i>)	starved ring, about 100 MHz/V (<i>est.</i>); inverter delay $1.66\times$ shorter at ff than tt (Liberty)
Loop bandwidth	$f_n \approx 400 \text{ kHz}$ at $PN = 16$; crossover $\approx 490 \text{ kHz}$ ($f_{\text{ref}}/13$) (<i>est.</i>)	crossover below $f_{\text{ref}}/10$ except $N = 1$ under 8.5 MHz ($f_{\text{ref}}/7.4$ at 6.25 MHz); Gardner limit [1] holds, $2\times$ margin
Loop filter	$I_{\text{cp}} \approx 20 \mu\text{A}$, $C_1 \approx 20 \text{ pF}$, $R \approx 22 \text{ k}\Omega$, $C_2 \approx C_1/10$ (<i>est.</i>)	(1) and (2) at $PN = 16$; phase margin $\approx 47^\circ$ (<i>est.</i>)
Lock time	under $50 \mu\text{s}$ (<i>est.</i>)	settling $4/(\zeta\omega_n) \approx 3 \mu\text{s}$ at $PN = 16$, $7 \mu\text{s}$ at $PN = 40$, plus acquisition
Jitter	simulated per corner; scope bound on silicon	absolute jitter is not measurable through the pads
Observed output	VCO $\div 8$ or $\div 16$ on uo_out[0]: 3.1 to 25 MHz (42 MHz fast)	pads limit an output to tens of MHz

Hand estimates of 24 September 2026, not yet simulated; ngspice corner results replace them at the Nov 5 evaluation. Section 2.1 uses the Liberty tables of gf180mcu_fd_sc_mcu7t5v0 (dffq_1, inv_1) at tt_025C_3v30 and ff_n40C_3v60.

2.1 Prescaler and frequency plan

Design constraint. The maximum clock frequency for synthesized logic on the chip is estimated at 50 MHz, so only the ripple prescaler runs faster. The divider, the PFD, the lock detector and the $\div 16$ observation flop all run at 50 MHz or below, at most 41.6 MHz at the fast corner.

Ratio. The smallest power of two that keeps the prescaler output at or below 50 MHz at the fast corner is 8: $333/2 = 166$ MHz, $333/4 = 83$ MHz and $333/8 = 41.6$ MHz. A $\div 4$ would be enough at the typical 200 MHz maximum (exactly 50 MHz) but not at the fast corner, so the design uses $\div 8$.

Toggle-flop speed. Each stage is a `gf180mcu_fd_sc_mcu7t5v0__dffq_1` with an `inv_1` from Q back to D. From the Liberty tables (clock slew 0.15 ns, about 13 fF on Q), the toggle loop is clk-to-Q plus inverter plus setup (Table 2). The first stage sees at most 333 MHz at the fast corner, a 3.0 ns period and 1.5 ns half-period: $3.2\times$ the 0.94 ns loop and $3.2\times$ the 0.47 ns minimum pulse width. Typical-corner cells at 333 MHz keep $1.8\times$. The three stage outputs are at most 166, 83 and 41.6 MHz.

TABLE 2 the prescaler's toggle loop from the Liberty tables

CORNER	CLK-TO-Q	INVERTER	SETUP	LOOP	LIBERTY MIN PERIOD
tt_025C_3v30	1.18 ns	0.13 ns	0.39 ns	1.70 ns (590 MHz)	1.71 ns
ff_n40C_3v60	0.68 ns	0.08 ns	0.18 ns	0.94 ns (1.06 GHz)	0.93 ns

Lock table. The loop locks where $50 \text{ MHz} \leq 8Nf_{\text{ref}} \leq 200 \text{ MHz}$, the typical VCO range (Table 3). Because $f_{\text{vco}} = 8Nf_{\text{ref}}$ is continuous in f_{ref} , $N = 1$ to 5 over 5 to 12.5 MHz covers 40 to 500 MHz without gaps, so any VCO band a corner produces inside that span still has a lock point.

TABLE 3 which N lock at each reference

f_{ref}	N THAT LOCK	f_{vco}	STEP $8f_{\text{ref}}$
5 MHz	2, 3, 4, 5	80, 120, 160, 200 MHz	40 MHz
6.25 MHz	1, 2, 3, 4	50, 100, 150, 200 MHz	50 MHz
10 MHz	1, 2	80, 160 MHz	80 MHz
12.5 MHz	1, 2	100, 200 MHz	100 MHz

2.2 Loop design

With the pump current I_{cp} , the VCO gain K_{vco} in rad/s/V, the total divide ratio PN and the series R and C_1 of the filter, the loop's natural frequency and damping are

$$\omega_n = \sqrt{\frac{I_{cp} K_{vco}}{2\pi PN C_1}}, \quad (1)$$

$$\zeta = \frac{RC_1 \omega_n}{2}. \quad (2)$$

With $K_{vco} = 2\pi \cdot 100 \text{ MHz/V}$, the values of Table 1 give $\omega_n \approx 2\pi \cdot 400 \text{ kHz}$ and $\zeta \approx 0.55$ at the nominal $PN = 16$. The zero is at 360 kHz and C_2 adds a pole at 4 MHz, so the phase margin is $\approx 47^\circ$ (*est.*). Both ω_n and ζ scale as $\sqrt{I_{cp}/PN}$: across $PN = 8$ to 40, ζ runs from 0.78 to 0.35 and the phase margin with C_2 from 55° to 33° (*est.*). I_{cp} stays at 20 μA ; doubling it with the trim at $PN = 32$ to 40 lifts ζ to 0.55 to 0.49 (margin 47° to 43°).

For acquisition, V_{ctrl} slews at most $I_{cp}/(C_1 + C_2) \approx 0.9 \text{ V}/\mu\text{s}$, so a 1.5 V swing takes a few microseconds; with the settling above, this sets the 50 μs lock-time bound.

Filter area. A 20 pF capacitor is large. At an assumed MIM density of 1 to 2 fF/ μm^2 it needs roughly 10,000 to 20,000 μm^2 , a large share of a two-tile area; this is also why the alternative implementation places the filter off chip.

Reference spur. Pump mismatch sets the reference spur. With a PFD reset pulse of about 1 ns (*est.*), a 10 % UP/DN mismatch gives a static phase offset of about 0.1 ns (0.23° at 6.25 MHz) and a ripple on V_{ctrl} every reference cycle. The C_2 pole at 4 MHz attenuates a 6.25 MHz spur by only about 5 dB, so $C_2 = C_1/10$ does little for spurs. The spur level is reported from ngspice; it cannot be measured through the pads.

Pad loading. The V_{ctrl} node is brought out on ua[0] for observation. The pad path (up to 5 pF [2]) sits in parallel with C_2 , so the on-chip C_2 is sized with the pad included: 2 pF plus 5 pF of pad drops the margin from 47° to 32° at $PN = 16$ and 33° to 23° at $PN = 40$ (*est.*).

WHY VCTRL IS NOT PROBED DIRECTLY

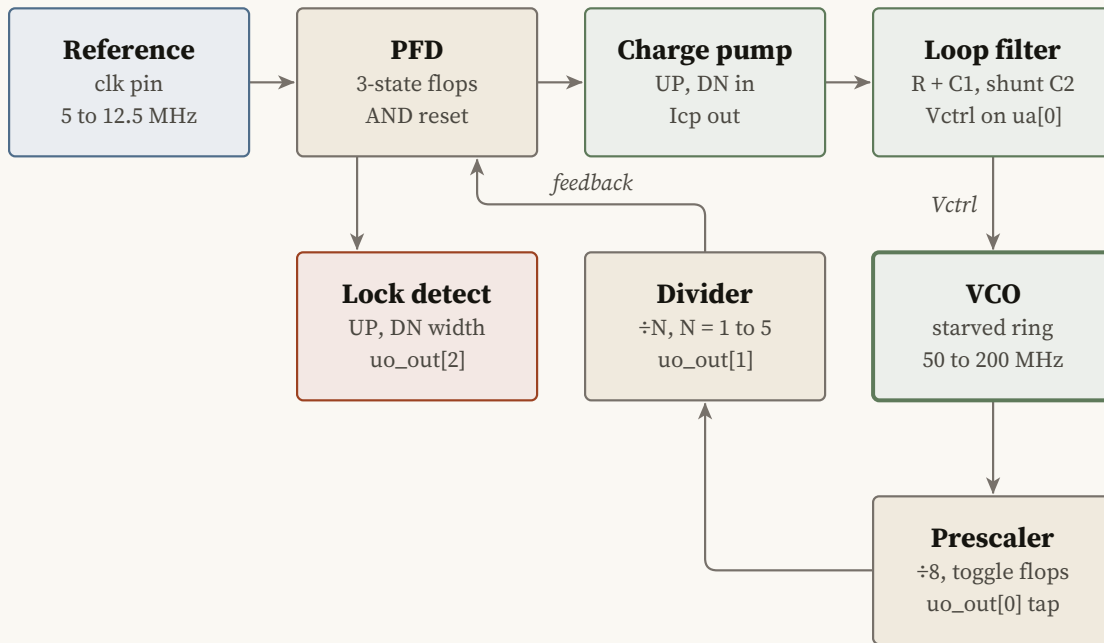
A scope probe (about 15 pF) on ua[0] would push the phase margin below 20° , so V_{ctrl} is observed only through a high-impedance buffer or with the loop open. Pad capacitance adds to C_2 , not C_1 ; a C_1 that is too small needs the R to C_1 node on a third analog pin.

Equations (1) and (2) with the Table 1 values, hand estimates of 24 September 2026; ngspice AC and transient results replace them at the Nov 5 evaluation.

3 Architecture

One loop runs from the reference through the PFD, pump and filter to the VCO, and back through the prescaler and divider. The lock detector watches the PFD, and the prescaler brings the VCO out to a pin.

FIGURE 1 *the charge-pump PLL on the analog tile*



Sand boxes are digital blocks built from standard cells, sage boxes are analog, the slate box meets a pin, and the rust box is the lock check. The heavy outline marks the VCO. The PFD drives the pump with UP and DN; the pump sources I_{cp} into the filter, whose node V_{ctrl} is also brought out on $ua[0]$. Only the prescaler runs above 50 MHz. Frequencies are estimates.

3.1 Blocks

*PF**D* (*digital*). Two D flops clocked by the reference and the feedback, with an AND reset. A short reset delay removes the dead zone. It is built from standard cells, with keep attributes on the delay chain so synthesis does not remove it. Its width is checked in ngspice, not in RTL. Its feedback flop and reset path see only the divided feedback, at most 41.6 MHz ($N = 1$, fast corner, during acquisition).

Charge pump (*analog*). Switched PMOS and NMOS current sources, with I_{cp} set by a mirror from a bias. The design concerns are UP/DN current mismatch across V_{ctrl} , which causes a static phase offset and reference spurs, and charge sharing at switching. A unity-gain buffer to equalise the switch nodes is optional and goes in only if ngspice shows it is needed.

Loop filter (*analog*). A series R and C_1 with shunt C_2 on the V_{ctrl} node, brought out on `ua[0]`; its sizing with the pad included is in Section 2.2.

VCO (*analog*). A current-starved inverter ring with an odd number of stages. V_{ctrl} sets the starving current. It is designed for monotonic tuning across corners, and the stage count is chosen from the ngspice sweep.

Prescaler (*digital*). Three ripple toggle flops (`dffq_1` with an inverter from Q to D) divide the VCO by 8. It is the only logic above 50 MHz, and it is placed next to the VCO output.

Divider and observation (*digital*). A programmable $\div N$ counter ($N = 1$ to 5; $N = 1$ passes the prescaler output through), clocked by the prescaler output at no more than 41.6 MHz. The observation output is the prescaler output ($VCO \div 8$) or one more toggle flop ($\div 16$), selected by `ui_in[7]`. At lock, $VCO \div 8$ is Nf_{ref} and the feedback output on `uo_out[1]` is f_{ref} , so the ratio of the two pins reads N .

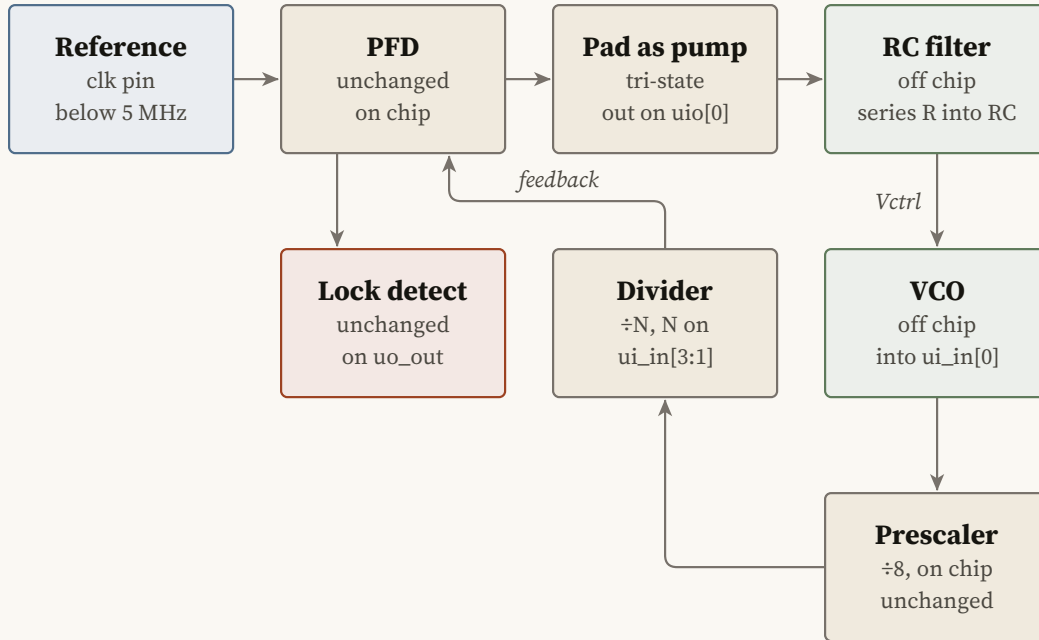
Lock detector (*digital*). Lock asserts when UP and DN pulses have stayed shorter than a threshold for K consecutive reference cycles, and drops when they do not. It is clocked by the reference.

3.2 Alternative implementation: digital tile, off-chip filter

The same loop can be built on the standard digital template. `uio[0]` acts as the charge pump: it is driven high on UP, low on DN and tri-stated (`uio_oe = 0`) otherwise, the same arrangement as the three-state phase comparator in a CD4046 [3]. During the PFD reset overlap, when UP and DN are both high, the pin stays tri-stated. An off-chip series resistor into an off-chip RC loop filter turns the pad into an approximate current source, with $I \approx (V_{DD} - V_{ctrl})/R$ on UP and V_{ctrl}/R on DN. The two match only at $V_{ctrl} = V_{DD}/2$, and the current varies with V_{ctrl} , so loop gain varies across the tuning range.

An off-chip VCO closes the loop through a `ui` input: a 4046-class part or a discrete oscillator in the low tens of MHz. The on-chip prescaler, $\div N$ divider, PFD and lock detector are unchanged, so the loop still multiplies by $8N$ and the core stays at or below 50 MHz whatever the VCO. With a VCO of about 30 MHz, $N = 5$ ($PN = 40$) needs a reference of about 0.75 MHz, so this implementation runs `clk` well below the 5 to 12.5 MHz of Table 1.

FIGURE 2 the alternative implementation, with the filter and VCO off chip



Sand boxes are digital blocks on the die and sage boxes are off-chip parts on the board; the slate box is the reference pin and the rust box is the lock check. The pump leaves the die on `uio[0]`, and the VCO returns on `ui_in[0]` into the on-chip prescaler.

In this form the silicon holds the digital half of the loop, as an integer-N synthesizer IC with an external VCO and loop filter does, but with a voltage-output pump, so it does not lock without the off-chip parts. A middle form keeps an analog tile for the VCO alone (one analog pin for V_{ctrl}) and puts the pump on `uio[0]` and the filter off chip.

Figure 2 from figures/cppll-alt.json, rendered with diagram-maker, 24 September 2026.

4 Interface and pin allocation

TABLE 4 *every pin of the analog-tile implementation*

PIN	DIR	USE
ui_in[2:0]	in	N (1 to 5; codes 0, 6 and 7 not supported)
ui_in[5:3]	in	spare
ui_in[6]	in	PLL enable (power-down of pump and VCO)
ui_in[7]	in	observation select ($\div 8$ or $\div 16$)
uo_out[0]	out	$VCO \div 8$ or $\div 16$
uo_out[1]	out	feedback divider output
uo_out[2]	out	lock
uo_out[3], uo_out[4]	out	UP, DN (debug; short pulses may not survive the pads)
uo_out[7:5]	out	driven low, spare
uio[1:0]	in	charge-pump current trim
uio[7:2]	in	spare
ua[0]	analog	V_{ctrl} , the loop filter node
ua[1]	analog	bias reference (external resistor), optional
clk, rst_n, ena	in	reference; standard

Alternative implementation. ui_in[0] external VCO in, ui_in[3:1] N , ui_in[6:4] spare, ui_in[7] enable; uio[0] pump out (dynamic uio_oe); uio[1] observation select (input); uo_out shows the feedback divider, lock, UP, DN and $VCO \div 8$ or $\div 16$.

5 Verification plan

5.1 CocoTB, digital blocks

The bench checks the PFD's UP/DN widths against known phase offsets, and both frequency-detection directions. The dead-zone reset pulse has no width in zero-delay RTL; CocoTB checks that it happens, and its width comes from ngspice. The bench checks the prescaler's $\div 8$ and the $\div 16$ tap, $\div N$ for every N from 1 to 5 (so the total ratio $8N$), and the lock detector's assert and deassert thresholds.

5.2 CocoTB, loop behaviour

Under ``ifdef SIM`, a real-number Verilog model stands in for the pump, filter and VCO. It integrates I_{cp} into the filter state each time step and sets the VCO period from V_{ctrl} , with K_{vco} and the range taken per corner from ngspice, up to the 333 MHz fast-corner maximum. For the alternative implementation, Python models the off-chip RC and VCO from `uio_oe/uio_out` and drives `ui_in[0]`. Lock is checked three ways, at every (f_{ref}, N) point of Table 3:

1. The lock flag rises within the bound.
2. Python edge timestamps show the average feedback period equal to the reference period, and the VCO period equal to the reference period divided by $8N$, with the phase error bounded.
3. The modelled V_{ctrl} settles.

The loop must also relock after a change of N . The bench is mutation-tested with a swapped UP/DN, a missing PFD reset, a divider off by one and a prescaler of $\div 4$.

5.3 Timing

Static timing constrains the prescaler output as a generated clock ($VCO \div 8$) at 50 MHz, and everything after it (the divider, the PFD feedback flop and the $\div 16$ flop) meets that. The reference domain (the PFD reference flop and the lock detector) is constrained at 12.5 MHz. The three prescaler flops are checked against the Liberty minimum period and pulse widths at the fast-corner VCO rate.

5.4 ngspice, analog blocks

PDK device models across corners (typical, fast, slow), supply $\pm 10\%$ and -40 to 125°C . The analysis covers:

- charge pump: UP/DN matching against V_{ctrl} , and charge sharing;
- loop filter: AC response and phase margin;
- VCO: frequency against V_{ctrl} , K_{vco} , range and monotonicity, plus period jitter from transient noise, injected with `trnoise` sources [4] sized from the device noise models (an estimate);
- prescaler: the first toggle flop driven by the VCO at the fast-corner maximum;
- full loop: a transient with the PFD, prescaler and divider built from the standard-cell SPICE netlists, showing V_{ctrl} settling and lock.

5.5 Layout checks

Layout is drawn in Magic or with a parameterised generator. Magic and KLayout DRC run on every revision; because precheck runs DRC only, netgen LVS is part of this plan, and the extracted parasitics are re-simulated before sign-off.

6 Schedule and milestones

TABLE 5 *what each course deadline delivers*

DATE	DELIVERABLE
Oct 8	Digital RTL (PFD, prescaler, divider, lock detector) and the real-number loop model, CocoTB green in CI; first-pass pump and VCO schematics; implementation fixed (analog tile or alternative)
Nov 5	Evaluation: ngspice corners for pump, filter, VCO and prescaler; full-loop transient lock; loop parameters fixed; layout under way
Nov 26	System verification: layout DRC-clean, LVS clean, post-layout re-simulation, Actions and precheck green, submission ready
Dec 3	Documentation and silicon measurement plan (V_{ctrl} against N , lock range, divided-output jitter)

7 Technical risks and mitigations

TABLE 6 *each design risk and what answers it*

RISK	MITIGATION
Layout errors not caught by precheck (DRC only)	netgen LVS and extracted re-simulation
Hand layout overruns the schedule	small topology, generator where possible, layout started by early November
Filter capacitor too large for the area	bring out the R to C_1 node for an off-chip C_1 (a third analog pin); capacitance on <code>ua[0]</code> only adds to C_2
Pump mismatch gives spurs or offset	sized from the ngspice sweep, current trim on <code>uio[1:0]</code>
VCO range misses at corners	the reference is adjustable, and $8Nf_{\text{ref}}$ covers 40 to 500 MHz without gaps
First prescaler flop too slow for the VCO	Liberty gives a 0.94 ns toggle loop at the fast corner against a 3.0 ns VCO period; confirmed in the ngspice prescaler transient
Loop dynamics vary with PN : ζ drops from 0.78 at $PN = 8$ to about 0.35 at $PN = 40$ (ω_n and ζ scale as $\sqrt{I_{\text{cp}}/PN}$), and with C_2 the phase margin falls from 55° to about 33° (<i>est.</i>)	I_{cp} stays at $20\ \mu\text{A}$; doubling it with the trim on <code>uio[1:0]</code> at $PN = 32$ to 40 lifts ζ to 0.55 to 0.49; loop parameters fixed in the Nov 5 evaluation
Standard cells inside the analog tile	digital blocks hardened as a macro and placed by hand
Pads limit observation	divided outputs only

References

- 1 F. M. Gardner, “Charge-pump phase-lock loops,” *IEEE Transactions on Communications*, vol. 28, no. 11, pp. 1849–1859, Nov. 1980.
- 2 Tiny Tapeout, “Analog specifications.” <https://tinytapeout.com/specs/analog/>
- 3 Texas Instruments, *CD4046B CMOS Micropower Phase-Locked Loop*, datasheet.
- 4 ngspice User’s Manual, independent sources (trnoise). <https://ngspice.sourceforge.io/docs.html>
- 5 Tiny Tapeout, ttgf-analog-template. <https://github.com/TinyTapeout/ttgf-analog-template>